Apostolos Foivos Varelas



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Profile

Software engineer with a background in artificial intelligence, robotics, and production Python systems. Experience spans autonomous navigation, reinforcement learning, computer vision, LLM integration, and geospatial data processing. Combines research-oriented problem solving with practical software engineering to build reliable intelligent systems for real-world applications.

Technical Skills

- **Programming:** Python, C
- **Robotics:** ROS 2, MuJoCo, autonomous navigation, task planning
- **Machine Learning:** PyTorch, scikit-learn, computer vision, deep learning, reinforcement learning, transformers, LLM integration
- **Data & Geospatial:** NumPy, pandas, OpenCV, GDAL, Rasterio, GeoPandas, xarray, Dask, Sentinel-1/2/3
- **Engineering:** Docker, Git, Linux, pytest, REST APIs, S3-compatible storage

Work Experience

Software Developer
Feb 2024 – Present

Eagle Projects — Perugia, Italy

- Won Eagle Projects' 2026 LLM + ROS 2 Robotics Challenge as part of a two-person team, developing an LLM-driven mission workflow with task planning, reusable robotic skills, and explicit safety constraints; awarded company-sponsored attendance at ROSCon Global 2026 in Toronto.
- Built a modular Sentinel-2 processing system that orchestrates reusable components for product acquisition, cloud filtering, multi-band raster generation, super-resolution, spectral indices, configurable land-cover classification, change detection, vector export, and web-map tiles.
- Developed an automated Sentinel-1 SBAS InSAR workflow for time-series ground-deformation analysis and DEM generation, producing GeoTIFF, GeoJSON, and CSV deliverables.
- Created agricultural and environmental decision-support services combining satellite, terrain, weather, backscatter, and air-quality data to estimate biomass, flood risk, drought, heat stress, soil erosion, soil moisture, land subsidence, and vegetation health.
- Developed an irrigation-support pipeline using Sentinel-1/2/3 and weather data to estimate daily water requirements and generate 72-hour forecasts; introduced gated ML inference with deterministic fallbacks and data lineage.
- Productionized the services with Docker, configuration-driven execution, unit tests, automated PDF reports, API uploads, and S3 synchronization with local fallback handling.

Research & Projects

M.Sc. Thesis — Safety-Constrained Autonomous Quadruped Navigation in Three-Dimensional Terrain 2026

- Investigated autonomous quadruped navigation in 3D terrain with an emphasis on enforcing safety constraints throughout the navigation process.

PyMARLzoo+ — Open-Source Multi-Agent Reinforcement Learning GitHub

- Co-integrated the MAVEN algorithm into PyMARLzoo+, an open-source framework for benchmarking cooperative multi-agent reinforcement-learning algorithms across complex environments.
- Ran experiments and reported evaluation results for the MAVEN integration; the broader PyMARLzoo+ framework was presented at AAMAS 2025.

B.Sc. Thesis — Reinforcement Learning in a Competitive Environment GitHub

- Created a two-agent soccer environment on a hexagonal grid and implemented tabular Q-learning and model-based prioritized sweeping with ϵ-greedy policies.
- Trained agents through self-play and recorded approximately 16,000 episodes and 874,000 transitions per agent for analysis and reproducibility.

Deepfake Video Detection — Deep Learning and Explainability GitHub

- Built an end-to-end PyTorch video-classification pipeline comparing ResNet50, Xception with squeeze-and-excitation, and Swin Transformer on 400 FaceForensics++ videos.
- Implemented face detection, mixed-precision training, video-level aggregation, extensive evaluation, and Grad-CAM analysis across face-crop and full-frame experiments.

Education

University of Piraeus — M.Sc. in Artificial Intelligence 2024 – 2026
Thesis: Safety-Constrained Autonomous Quadruped Navigation in Three-Dimensional Terrain

University of Piraeus — B.Sc. in Digital Systems / Computer Science 2020 – 2024
Thesis: Reinforcement Learning in a Competitive Environment

Languages

Greek: Native **English:** C2 Proficiency (Michigan ECPE)